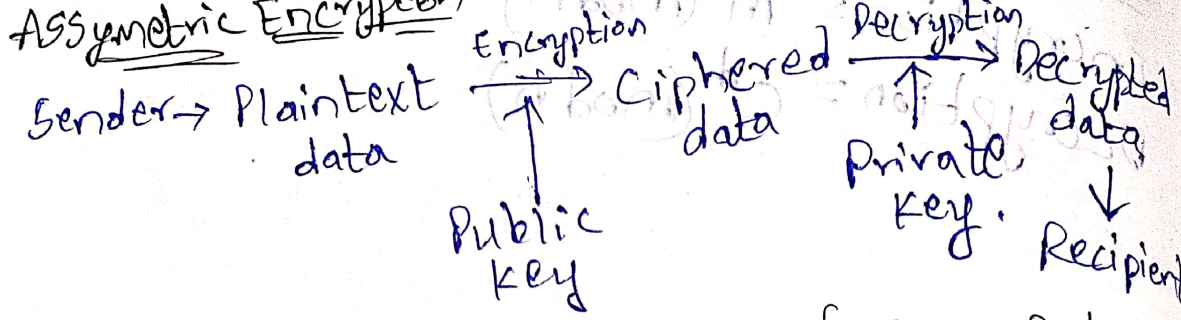
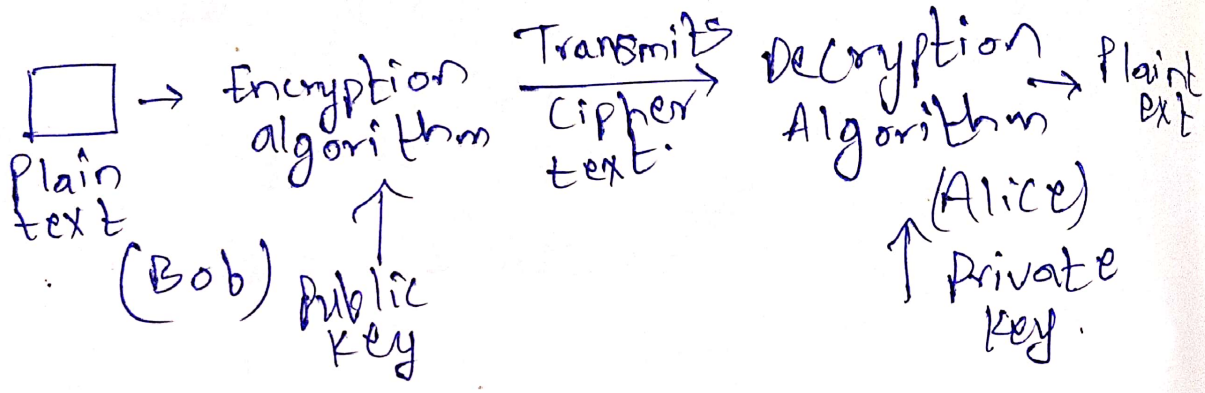
CO-3

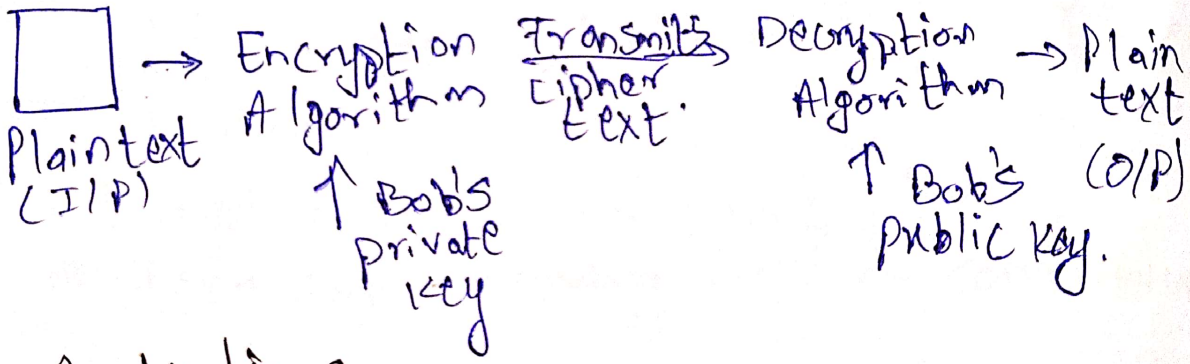
Assymetric Encryption



Public key Cryptosystems for confidentiality



For Authentication



Applications

→ Encryption /decryption

→ Digital Signature.

→ Key exchange.

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RSA

* Based on Prime numbers, Exponentiation & factorization.
* Two exponents ⇒ e, d → decryption.
  ↓ encryption

Plain text ⇒ M
Cipher text ⇒ C

$C = M^e \% n$ ⇒ Encryption.

$M = C^d \% n$ ⇒ Decryption.

Public key = (e, n)
Private key = (d, n)

→ P & Q = Prime numbers.

$n = P \times Q$.

Euler's Totient Function $\phi(n) = (p-1) \times (q-1)$

Ex:- P = 17, q = 11

$n = p \times q = 17 \times 11 = 187$

$\phi(n) = (17-1) \times (11-1) = 16 \times 10 = 160$

$1 < e < \phi(n)$

$1 < e < 160$

e = 7. should be coprime.

$d \times e \% n = 1$

$d \times 7 \% 160 = 1$

$23 \times 7 \% 160 = 1$

$161 \% 160 = 1$.

e = 7

d = 23

m = 88

$c = 88^7 \% 187$ $(m^e \% n)$

c = 11

$m = c^d \% n$

$= 11^{23} \% 187$

= 88

RSA Algorithm attacks

→ Brute force key search

→ Mathematical attacks.

→ Chosen ciphertext attacks.

→ Timing attacks.

Ex:- p = 13, q = 17

n = p x q

n = 221

$\phi(n) = 12 \times 16$

= 192

d = 77, e = 5

Public key (e, n) = (5, 221)

Private key (d, n) = (77, 221)

$c = m^e \% n$

$= 43^5 \% 221$

= 127

$m = c^d \% n$

$= 127^{77} \% 221$

= 43.

p = 13, q = 17, e = 11, d = 7

n = p × q

= 221

$\phi(n) = 12 \times 16$

$= 192$

d × e % 192 = 1

Diffie Hellman Algorithm

→ p = Large Prime

g = Generator

→ secret a → keep private ← secret b

→ $A = g^a \% p$

$B = g^b \% p$

→ $S = B^a \% p$

$S = A^b \% p$

Ex:- p = 23

g = 5

a = 6

b = 15

$A = g^a \% p$

$= 5^6 \% 23$

$= 15625 \% 23 = 8$

$B = g^b \% p$

$= 5^{15} \% 23$

$= 19515 \% 23 = 19$

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Alice Sends 8 to Bob.
Bob sends 19 to Alice.

$S = B^a \% P$
$S = 19^6 \% 23$
$S = 2$

$S = A^b \% P$
$S = 8^{19} \% 23$
$S = 2$

* Final Shared secret = 2

Elgamal Cryptosystem

→ P = large prime number
→ g = primenumber
→ A, $(1 < A < P-2)$ & compute $gA = g^A \% P$
→ Publickey (P, g, gA)
→ Private key is A.

Ex:- P = 11
g = 7
A = 3
$gA = 7^{**}3 \% 11 = 2$
Public key = (11, 7, 2)
Private key = 3.

Ex:- M=8, K=4

$C_1 = g^K \% P$

$= 7^4 \% 11 = 3$

$C_2 = (M \times g^{A^K}) \% P$

$= (8 \times 2^4) \% P$

$= (8 \times 16) \% 11$

$= 128 \% 11$

$= 7$

Cipher Text = (3,7)

Ex:- P=23, g=7, A=11, M=13, K=6

$gA = 7^{11} \% 23$

$= 22$

Public key = (23, 7, 22)

Private key = 11

$C_1 = g^K \% P$

$= 7^6 \% 23$

$= 4$

$C_2 = (M \cdot g^{A^K} \% P)$

$= 13 \cdot 22^6 \% 23$

$= 13$

$(C_1, C_2) = (4, 13)$

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Elliptic Curve Cryptosystem

1) Closure
2) Associativity.
3) Commutativity.
4) Existence of identity.
5) Existence of inverise.

$p = 23, a = 1, b = 1$

$E :- y^2 = x^3 + x + 1 \ \% 23$

$4a^3 + 27b^2 = 8 \ \% 23$

$4(1)^3 + 27(1)^2 \ \% 23 = 8 \ \% 23$

$31 \ \% 23 = 8 \ \% 23$

$8 = 8$

General equation of elliptic curve,

$$y^2 + b_1xy + b_2y = x^3 + a_1x^2 + a_2x + a_3$$

(or)

$$y^2 = x^3 + ax + b.$$

Elliptic Curve Cryptography Algorithm

1) Choose $E(a, b)$ with an elliptic curve over $GF(p)$
2) Choose a point on the curve G as a generator point.
3) Choose and integer d.

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4) Calculate $e = dG$

5) Announces $(E(a,b), G, e)$ as public key. And keeps d as private key.

Encryption

$$C_1 = r \times G \qquad (\because e = dG)$$

$$C_2 = M + r \times e$$

$$C = (C_1, C_2).$$

Decryption

$$M = C_2 - d \times C_1.$$

Ex:-

E: $y^2 = x^3 + 2x + 2 \bmod 17$

$G = (5,1)$

$d = 7$

$e = 7G$

$7G$